

Final supplementary test 4

Total points **10/40** ?

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Hyunseo

✓ Q1. *

2/2

Sam has diabetes, a disease that makes his body unable to control the glucose level in his blood. Sam's doctor wants to show Sam which foods to avoid to keep his blood glucose level below 7.5.

For four days Sam recorded what he ate for lunch and measured his blood glucose level 1 hour after eating.

Day	Food eaten	Blood glucose level 1 hour after eating
1	bread with peanut butter	9.5
2	bread with chicken	8.4
3	crackers with peanut butter	7.3
4	crackers with chicken	6.9

Which food should Sam avoid eating the most?

(A) bread (B) crackers (C) chicken (D) peanut butter

☒ A☐ B☐ C☐ D

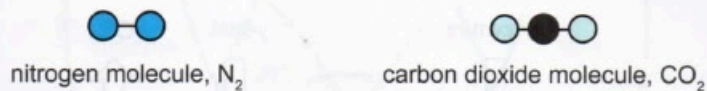
✗ Q2. *

0/2

An atom is the smallest unit making up matter. Different atoms can be represented like this:

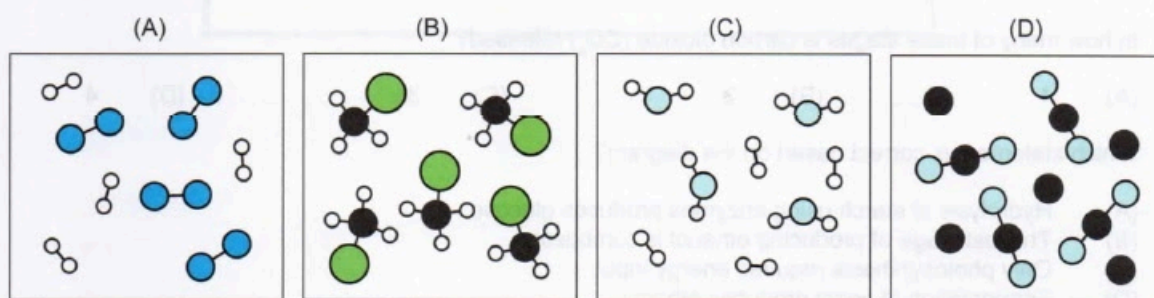


Atoms can join together to form molecules. For example



Pure substances contain only one type of atom or molecule.

Which of these diagrams represents a pure substance?

☒ A

✗

☐ B☐ C☐ D

Correct answer

☒ B

✓ Q3. *

2/2

Most adult mammals have four types of teeth:

- incisors for cutting food;
- canines for tearing food;
- premolars for crushing food;
- molars for grinding food.

Carnivores are meat-eating mammals. They mainly cut and tear meat and crush bones. They rarely grind their food. The table lists the types of teeth that four different adult mammals normally have.

Mammal	Number of teeth on one side of the top jaw			
	Incisors	Canines	Premolars	Molars
W	3	1	3	1
X	0	0	3	3
Y	3	1	2	4
Z	1	0	0	3

Which of the mammals listed in the table is most likely a carnivore?

(A) W (B) X (C) Y (D) Z

☒ A



☐ B

☐ C

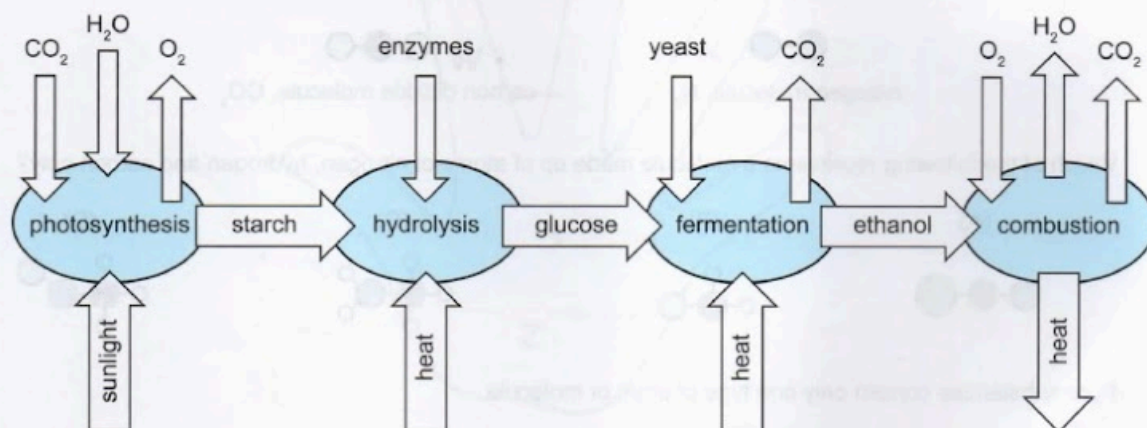
☐ D



✓ Q4. *

2/2

The diagram shows the four stages involved in making and using ethanol as a fuel.



Which statement is correct based on the diagram?

- (A) Hydrolysis of starch using enzymes produces glucose.
- (B) The last stage of producing ethanol is combustion.
- (C) Only photosynthesis requires energy input.
- (D) Fermentation of yeast produces ethanol.

☒ A



☐ B

☐ C

☐ D



✗ Q5. *

0/2

The table lists some of the components in 100 grams (g) of the milk that some mammals produce.

Mammal	Protein (g)	Fat (g)	Lactose (g)	Energy (kcal)
cow	3.3	3.9	4.6	66
human	1.3	4.1	7.2	69
goat	3.1	3.7	4.4	62
sheep	5.4	5.8	5.1	93

How many grams of protein are there in 200 g of goat's milk?

- (A) 3.1 (B) 3.3 (C) 6.2 (D) 6.6

☒ A

✗

☐ B

☐ C

☐ D

Correct answer

☒ C



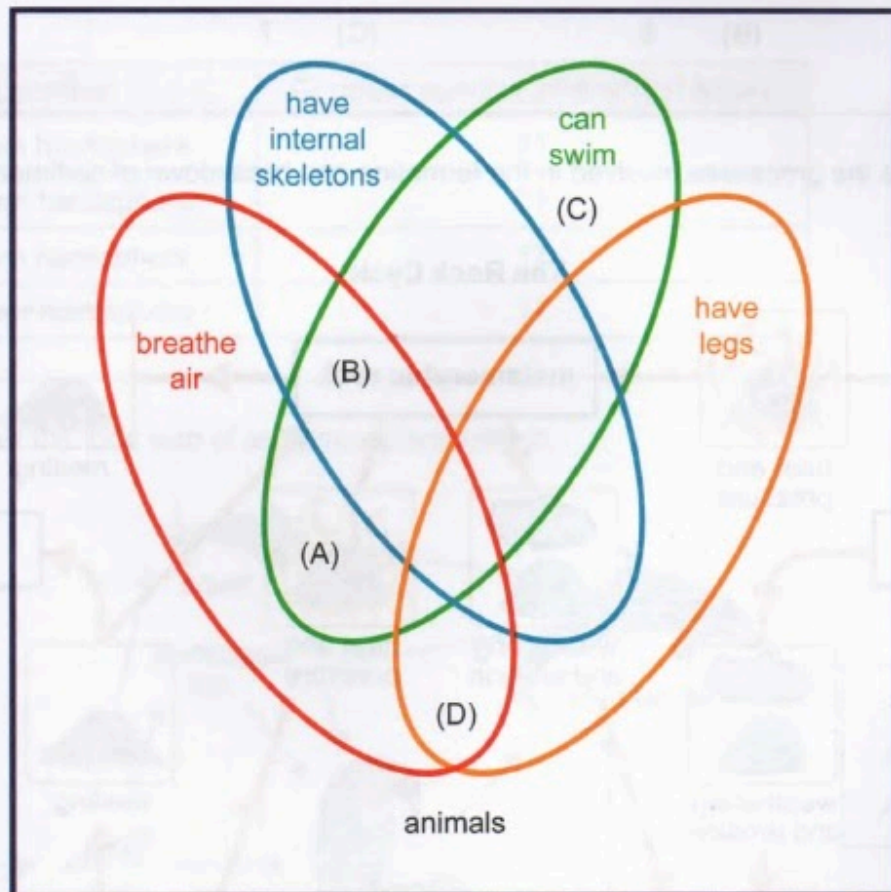
✓ Q6. *

2/2

This Venn diagram is used to group animals based on their physical characteristics.

An animal has no internal skeleton, no legs, it can swim and it breathes air.

In which area of the diagram does this animal belong?

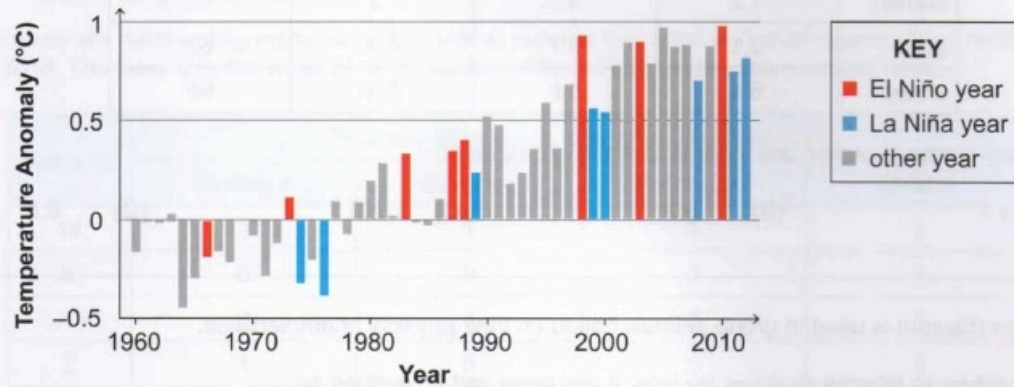
☒ A☐ B☐ C☐ D

X Q7.*

0/2

El Niño and La Niña are weather patterns resulting from periodic temperature changes in a part of the Pacific Ocean. These weather patterns influence the world's climate.

The graph shows the variation in global temperatures over a number of years. Changes from the baseline are called anomalies.



How many El Niño years occurred when the global temperature was above the baseline?

- (A) 13 (B) 8 (C) 7 (D) 6

☒ A

X

☐ B

☐ C

☐ D

Correct answer

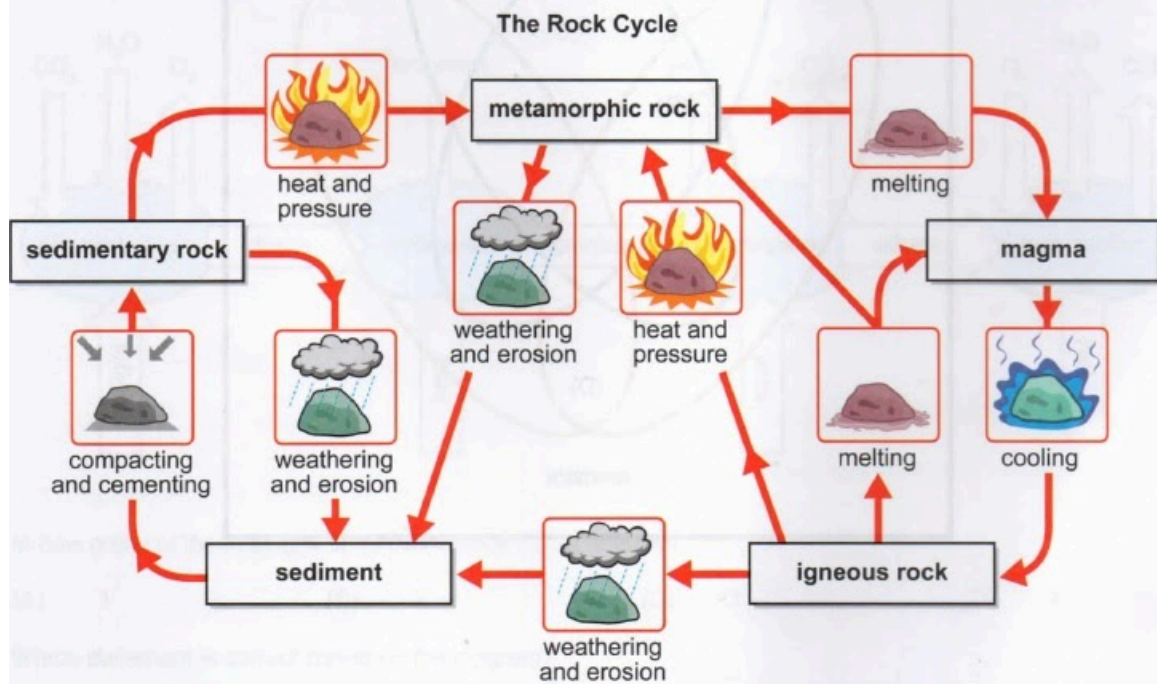
☒ C



X Q8. *

0/2

This diagram shows the processes involved in the formation and breakdown of sedimentary, metamorphic and igneous rocks.



Limestone is a sedimentary rock that can be changed into marble, a metamorphic rock.

What process(es) is involved in changing limestone into marble?

- | | |
|----------------------------|------------------------------|
| (A) melting | (B) heat and pressure |
| (C) weathering and erosion | (D) compacting and cementing |

☒ A

X

☐ B

☐ C

☐ D

Correct answer

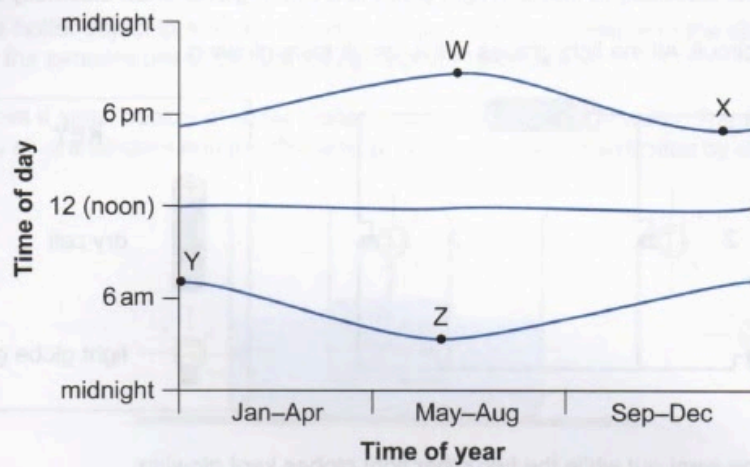
☒ B



X Q9. *

0/2

The graph shows the times of sunrise and sunset at one location on Earth over a year.



In summer the number of daylight hours is greater than it is in winter. In the middle of the year, it is summer in the northern hemisphere and winter in the southern hemisphere.

Which of these gives the location represented by the graph and the greatest number of daylight hours at this location?

	Location	Greatest number of daylight hours
(A)	northern hemisphere	21
(B)	southern hemisphere	21
(C)	northern hemisphere	17
(D)	southern hemisphere	17

☒ A

X

☐ B

☐ C

☐ D

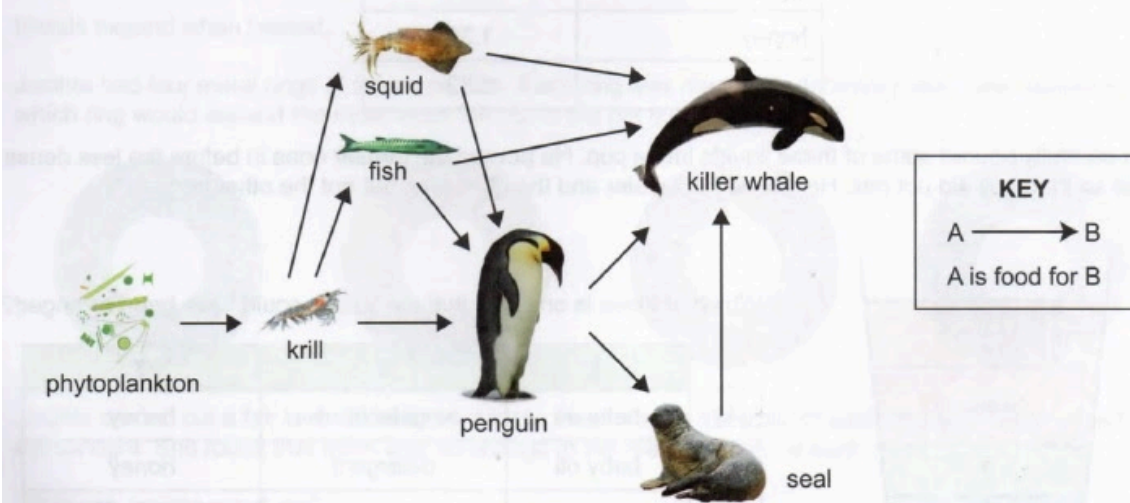
Correct answer

☒ C


✗ Q10. *

0/2

The diagram shows the food web of an Antarctic ecosystem.



A change in temperature has resulted in a rapid decrease in the number of krill.

Which of the following is most likely to happen?

- (A) The number of phytoplankton will decrease. (B) The number of penguins will increase.
(C) The number of seals will not be affected. (D) The number of squid will decrease.

☒ A

✗

☐ B

☐ C

☐ D

Correct answer

☒ D



X Q11. *

0/2

The table lists the densities of some common liquids. A liquid with a lower density will float on top of a liquid with a higher density.

Material	Density (g/cm ³)
water	1.00
vegetable oil	0.91
detergent	1.03
glycerine	1.26
honey	1.36
baby oil	0.82

Ben carefully poured some of these liquids into a cup. He poured the denser ones in before the less dense ones so that they did not mix. He labelled the water and the glycerine, but not the other layers.



Which of these is one way that the liquids could have been arranged?

	1	2	3
(A)	baby oil	vegetable oil	honey
(B)	baby oil	detergent	honey
(C)	vegetable oil	honey	detergent
(D)	honey	detergent	baby oil

☒ A

X

☐ B

☐ C

☐ D

Correct answer

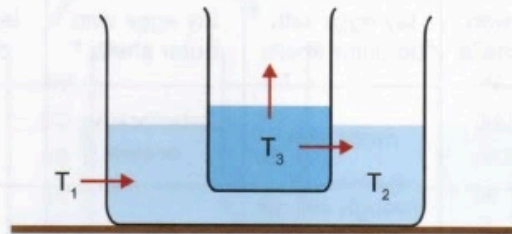
☒ B


✗ Q12. *

0/2

Heat flows from a hotter object to a cooler object in contact. The temperature of the object receiving heat will increase and the temperature of the one losing heat will decrease.

The diagram shows a small beaker of water placed in a large container of water. The temperatures (T) of the water and the air are labelled and the direction of heat movement is indicated by arrows.



The containers were left on the table for 24 hours.

What will be the values of T_1 , T_2 and T_3 ?

- (A) They will all be at the average of T_1 , T_2 and T_3 .
- (B) They will all be at the current room temperature.
- (C) T_1 will change to the average of T_2 and T_3 .
- (D) T_1 and T_3 will become the same as T_2 .

☒ A

✗

☐ B☐ C☐ D

Correct answer

☒ B

✓ Q13. *

2/2

Vertebrates are animals that have a backbone.

The table shows five different groups of vertebrates and their typical characteristics.

	Fish	Amphibians	Reptiles	Birds	Mammals
Form of reproduction	lay eggs with no outer shells	lay eggs with no outer shells	lay eggs with outer shells *	lay eggs with outer shells	give birth to live young **
Body covering	interlocking scales	moist skin	interlocking scales	feathers	dry skin with hair
Method of breathing	gills in head	through skin or gills; lungs as adult	lungs	lungs	lungs
Cold- or warm-blooded	cold-blooded	cold-blooded	cold-blooded	warm-blooded	warm-blooded

* Some snakes give birth to live young ** Echidnas and platypuses lay eggs

Which statement can be inferred from the information in the table?

- (A) Echidnas and platypuses are mammals.
- (B) Some snakes are reptiles and some are mammals.
- (C) Animals that lay eggs with outer shells are warm-blooded.
- (D) Amphibians breathe through their skin when they are on land.

☒ A

☐ B

☐ C

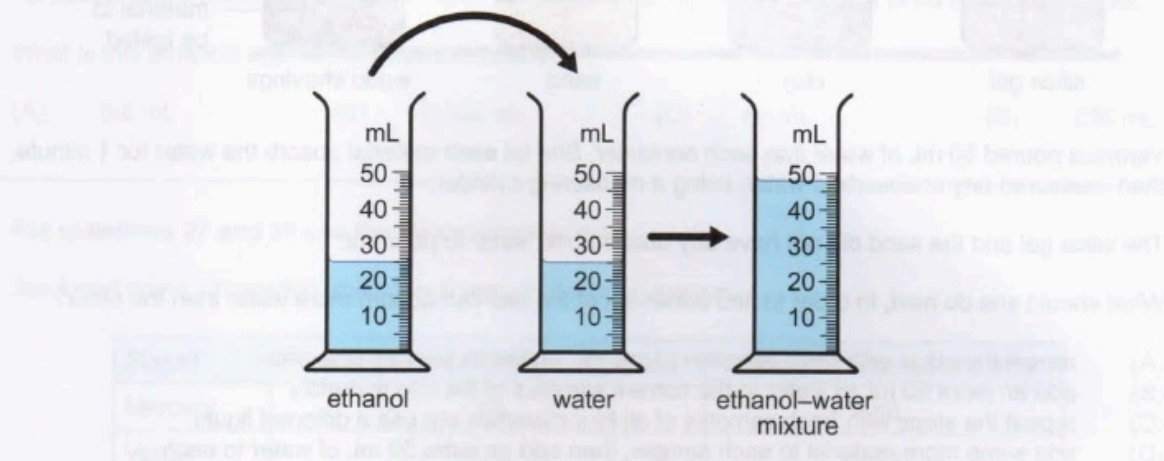
☐ D


✗ Q14. *

0/2

Ethanol and water are both colourless liquids. They mix well with each other.

Mary measured out 25 mL of ethanol and 25 mL of water using measuring cylinders. She then poured all the ethanol into the water cylinder without spilling. The diagram shows the volume of each liquid.



Which statement is supported by the observation when mixing the two liquids?

- (A) Mixing ethanol and water introduces gas into the mixture.
- (B) The scales of the two cylinders must be slightly different.
- (C) The total volume of the mixture is double the volume of each individual component.
- (D) The ethanol-water mixture takes up less space than the sum of the individual liquids.

☒ A

✗

☐ B

☐ C

☐ D

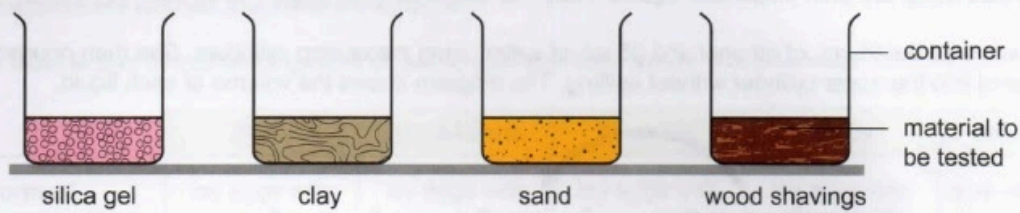
Correct answer

☒ D

X Q15. *

0/2

Veronica had four different materials. She wanted to find out which one is the most absorbent when using the same volume of each material.



Veronica poured 50 mL of water into each container. She let each material absorb the water for 1 minute, then measured any unabsorbed water, using a measuring cylinder.

The silica gel and the sand did not have any unabsorbed water to pour out.

What should she do next, in order to find out which of the two can absorb more water than the other?

- (A) repeat the steps with fresh samples of the two materials but use less water
- (B) add an extra 50 mL of water to the current samples of the two materials
- (C) repeat the steps with fresh samples of all four materials but use a different liquid
- (D) add some more material to each sample, then add an extra 50 mL of water to each

☒ A

X

☐ B☐ C☐ D

Correct answer

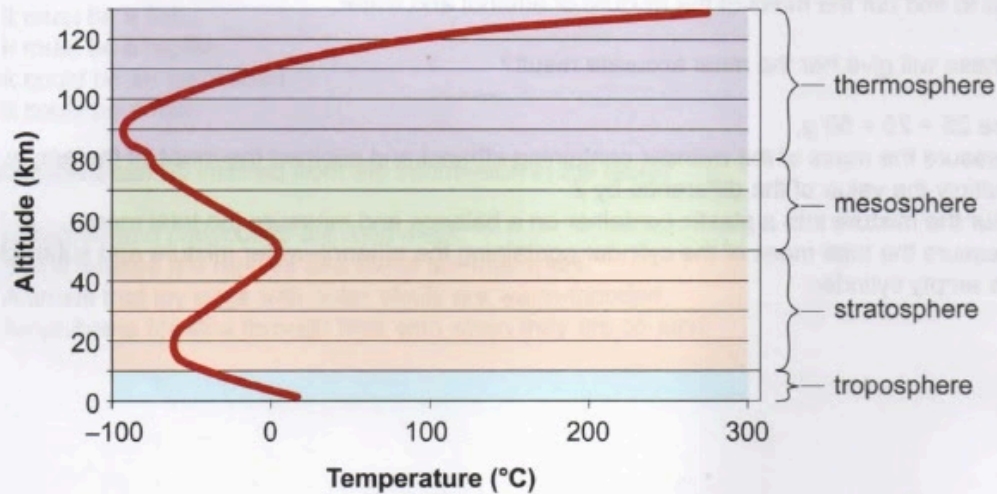
☒ B

✗ Q16. *

0/2

The air surrounding Earth is called the atmosphere. It consists of four layers: the troposphere, the stratosphere, the mesosphere and the thermosphere.

The graph shows the temperature of the atmosphere at different altitudes (height above sea level).



Which statement about the atmosphere is correct?

- (A) The greater the altitude, the higher the temperature.
- (B) The troposphere is the coldest part of the atmosphere.
- (C) The temperature drops when moving up through the mesosphere.
- (D) The temperature rises continuously when moving up through the stratosphere.

☒ A

✗

☐ B☐ C☐ D

Correct answer

☒ C

✗ Q17.*

0/2

Joe found some information about the planets in the solar system.

Planet	Distance from Sun (millions km)	Diameter (thousands km)
Mercury	58	4.88
Venus	108	12.10
Earth	150	12.76
Mars	228	6.78
Jupiter	778	139.82
Saturn	1429	116.46
Uranus	2871	50.72
Neptune	4504	49.25

Which statement is supported by the data in the table?

- (A) The further the planet is from the Sun, the colder the planet is.
- (B) The further the planet is from the Sun, the larger its diameter is.
- (C) As distance from the Sun increases, the distance between neighbouring planets gets greater.
- (D) The distance of each planet from the Sun is over 1000 times greater than the diameter of the planet.

☒ A

✗

☐ B

☐ C

☐ D

Correct answer

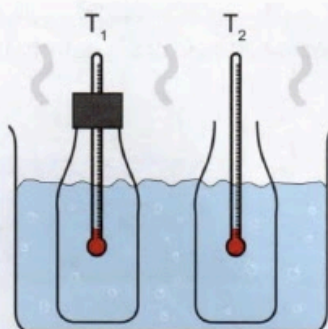
☒ D



✗ Q18. *

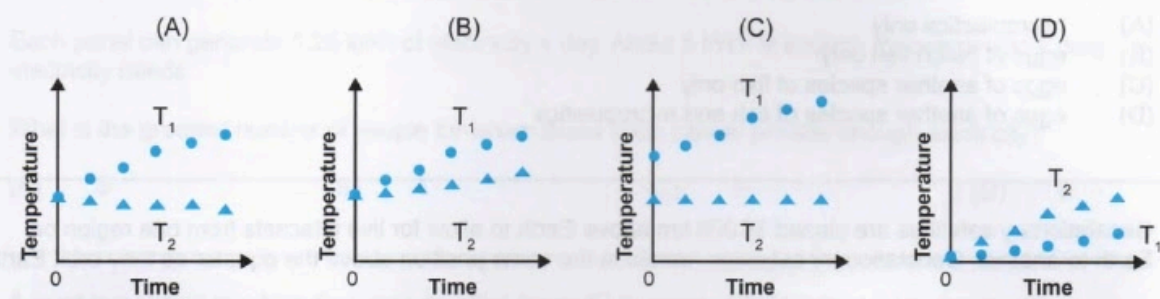
0/2

Ben had two identical empty glass bottles, one with a cap and the other without. He placed both bottles in hot water and recorded the temperatures (T) inside each bottle every minute for 5 minutes.



Ben then graphed his results.

Which graph correctly represents Ben's results?



☒ A

✗

☐ B

☐ C

☐ D

Correct answer

☒ B



✗ Q19. *

0/2

A group of students wanted to find out if the time it took for water to freeze depends on the volume of the water. Part of their report is shown below.

The effect of volume on freezing time

Aim: To determine if the volume of water affects how long it takes to freeze

Equipment: 250 mL measuring jug, 4 plastic cups, water, freezer, timer

Method:

1. Measure 50 mL of water in the measuring jug and transfer the water into a plastic cup.
2. Repeat step 1 so that the 2nd cup contains 100 mL, the 3rd cup contains 150 mL and the 4th cup contains 200 mL.
3. Place all four cups in the freezer.
4. Check each cup every 10 minutes and record the time when the water freezes.

Which of the following most significantly improves the accuracy of the test?

- (A) use a stopwatch to measure the time
- (B) use a measuring cylinder instead of a measuring jug to measure the volume of the water
- (C) check the cups in the freezer every 2 minutes instead of every 10 minutes
- (D) conduct the experiment for each cup five times and calculate the average of the results

☒ A

✗

☐ B

☐ C

☐ D

Correct answer

☒ C

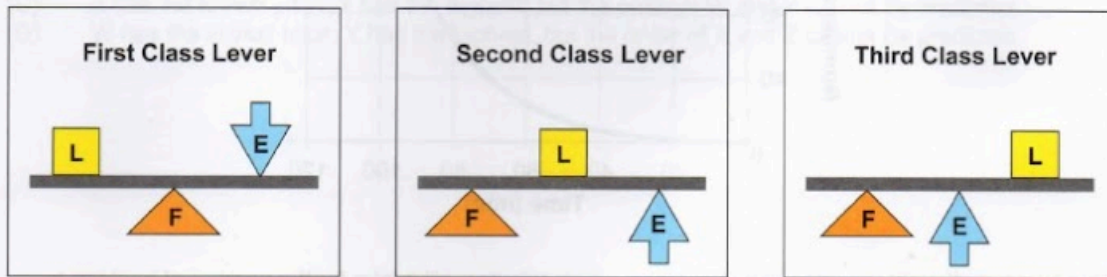


✗ Q20. *

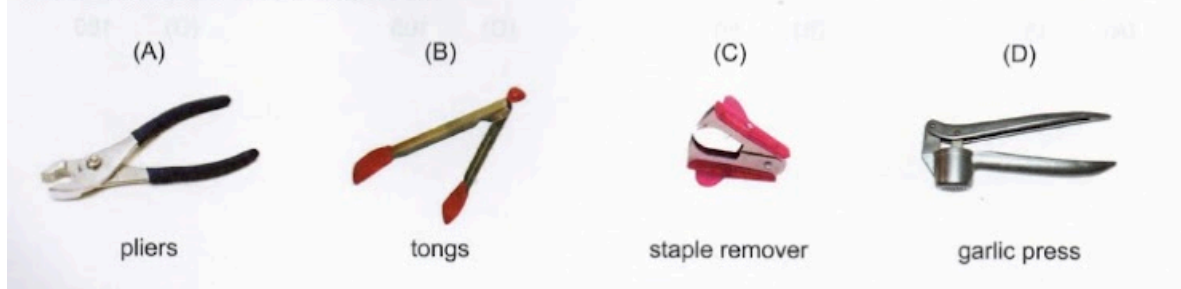
0/2

A lever is a simple machine that uses an effort force (E) to move or hold a load (L) by turning the lever about a pivot point, the fulcrum (F).

There are three classes of levers:



Which of these is a second class lever?



☒ A

✗

☐ B

☐ C

☐ D

Correct answer

☒ D

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